Exercise 1: Mr. M has to pay Mr. N 1,000 USD in the second year and 3,000 USD in the fifth year from the present time. If they renegotiate the contract to pay off the entire amount in the third year, how much does Mr. M need to pay? Assume an interest rate of 6% per year.

Solution

- Year 2 -> year 3: $N_3 * (1 + r) = 1000 * (1 + 0.06) = 1060$
- Year 5 -> year 3: $N_5 / (1+r)^2 = 3000 / (1 + 0.06)^2 \approx 2670.06$

⇨ Total in year 3: $1060 + 2670.06 \approx 3730.06$ USD

Exercise 2: A computer is sold for 2,000 USD with an additional 12-month installment plan of 250 USD per month, based on an 18% monthly interest rate. How much would it cost to buy the computer if paid in full immediately?

Solution

$$PV = PMT \times \frac{1-(1+r)^{-n}}{r}$$

$PV = PMT \times (1 - (1 + r)^{-n}) / r$

Where:

- $PMT = 250$
- $r = 0.18$
- $n = 12$

$PV = 250 * (1 - (1 + 0.18)^{-12}) / 0.18 \approx 1198.5$ USD

Exercise 3* Mr. A has made seven annual payments of 10 million VND at the end of each year for a loan of 100 million VND, with an interest rate of 5% per year. If he wants to pay off the remaining amount over the next five years, how much must he pay annually?

Hint: New loan principal = Previous principal + Interest - Payment (expand it and derive closed-form solution, see appendix for formula)

Solution

$$P_7 = P_0(1+r)^7 - PMT \times \frac{(1+r)^7 - 1}{r}$$

Remaining loan balance after 7 years: P7 = P0 × (1 + r)^7 − PMT × [(1 + r)^7 − 1] / r

Where:

- P0 = 100 (initial loan, million VND)
- r = 0.05
- PMT = 10

P7 = 100 × (1.05) ^7 − 10 × [(1.05)^7 − 1] / 0.05 ≈ 59.29

$$PMT = \frac{P \cdot r}{1 - (1+r)^{-n}}$$

New annual payment for next 5 years:

PMT = P × r / [1 − (1 + r)^(-n)]

Where:

- P = 59.29
- r = 0.05
- n = 5

PMT = 59.29 × 0.05 / [1 − (1.05)^(-5)] ≈13.69

Mr. A has to pay **13.69 million VND** for the next five years to pay off the remaining loan.

Exercise 4 You rent out a house for 6,000 USD per year, with payments made at the end of each year for five years. The entire rental income is deposited in a bank with a compound interest rate of 6% per year. After five years, what will be the total amount, including principal and interest?

Solution

Accumulated amount at the end of n periods is:

$$FV = PMT \times \frac{(1+r)^n - 1}{r}$$

FV = 6000 × [(1.06)^5 − 1] / 0.06 = 33,820 USD

total amount is **33,820 USD** after 5 years

Exercise 5 We are considering two projects, A and B, with 1,000 USD as initial investment cost in the first year (Year 0) for each project, and no other cost for other years (Year 1 to Year 4). The estimated benefits for each project are as follows:

Project	Year 1	Year 2	Year 3	Year 4
A	500	500	500	500
B	800	800	200	200

Calculate the Net Present Value (NPV) and Return on Investment (ROI) to determine which project should be selected. Assume an interest rate of 10%

Solution

$$NPV = \sum_{t=1}^{n} \frac{B_t - C_t}{(1+r)^t}$$

$$\text{ROI} = \text{NetProfit} / \text{InvestmentCost} * 100$$

Project A

NPV = 500/(1.1)^1 + 500/(1.1)^2 + 500/(1.1)^3 + 500/(1.1)^4 − 1000
= 454.55 + 413.22 + 375.66 + 341.51 − 1000
= 1584.94 − 1000 = **584.94 USD**

ROI = 584.94 / 1000 × 100% = 58.49%

Project B

NPV = 800/ (1.1) ^1 + 800/(1.1)^2 + 200/(1.1)^3 + 200/(1.1)^4 − 1000
= 727.27 + 661.16 + 150.26 + 136.60 − 1000
= 1675.29 − 1000 = 675.29 USD

ROI = 675.29 / 1000 × 100% = **67.53%**

Exercise 6 Two projects, A and B, have been estimated with the following data:

Project A	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Investment	500	0	100	0	0	0
Benefit	0	250	350	450	500	-100

Project B	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Investment	100	0	50	0	0	0
Benefit	0	-100	200	200	200	300

Given an interest rate of **10%**, calculate the **NPV** and **ROI** for each project.

Solution

$$NPV = \sum_{t=1}^{n} \frac{B_t - C_t}{(1+r)^t}$$

ROI = NetProfit / InvestmentCost * 100

Project A

Cash Flows (Benefit − Investment):

Year 0: $0 - 500 = -500$
Year 1: $250 - 0 = +250$
Year 2: $350 - 100 = +250$
Year 3: $450 - 0 = +450$
Year 4: $500 - 0 = +500$
Year 5: $-100 - 0 = -100$

NPV = -500 + 250/(1.1)^1 + 250/(1.1)^2 + 450/(1.1)^3 + 500/(1.1)^4 - 100/(1.1)^5
= -500 + 227.27 + 206.61 + 338.09 + 341.51 - 62.09
= 551.39 USD

ROI = 551.39 / 500 × 100 = 110.28%

Project B

Cash Flows (Benefit − Investment):

Year 0: $0 - 100 = -100$
Year 1: $-100 - 0 = -100$
Year 2: $200 - 50 = +150$
Year 3: $200 - 0 = +200$
Year 4: $200 - 0 = +200$
Year 5: $300 - 0 = +300$

NPV = -100 - 100/(1.1)^1 + 150/(1.1)^2 + 200/(1.1)^3 + 200/(1.1)^4 + 300/(1.1)^5
= -100 - 90.91 + 123.97 + 150.26 + 136.60 + 186.28
= 406.20 USD

ROI = 406.20 / 100 × 100 = 406.20%